Lung Cancer Detection System Using CNN

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Company Profile

- Accolade Tech Solutions Pvt. Ltd. is a well-regarded software and web development company headquartered in Mangalore, India.
- With extensive industry experience, deep technological expertise, and a broad range of services, Accolade Tech Solutions enables clients to enhance their business operations.
- The company offers comprehensive IT consulting services, guiding organizations on effectively leveraging technology to meet their business goals.

Introduction to the Project

Abstract

- This project presents a deep learning-based Lung Cancer Detection System designed to classify CT-scan images into benign, malignant, or normal categories using Convolutional Neural Networks (CNNs).

- The system addresses limitations in traditional diagnostic methods by providing automated, accurate, and real-time analysis of medical images. Integrated with a web-based interface, it allows patients to upload scans and receive immediate diagnostic feedback, while doctors can access detailed reports and manage patient records.

- The system is built using open-source tools including Flask, TensorFlow, and MySQL, making it cost-effective and scalable. Designed with a focus on usability, data security, and clinical relevance, this solution aims to enhance early cancer detection, streamline medical workflows, and support informed decision-making in healthcare environments.

SRS Details

❖ Objective:

To develop a machine learning-based web application that classifies lung CT-Scan images into benign, malignant, or normal categories, enabling early and accurate cancer detection with minimal manual intervention.

❖ Intended Users:

- Patients – Upload scans, receive results.
- Doctors – View predictions, provide medical notes.
- Admins – Manage users and oversee operations.

❖ Goals:

- Automate cancer detection using CNNs.
- Reduce diagnostic time and manual errors.
- Generate downloadable PDF reports.
- Provide secure, role-based access.
- Ensure compliance with healthcare data privacy.

SRS Details

❖ Functional Requirements:

- User Module: Registration, login, image upload, prediction view, PDF download.
- Doctor Module: View patient history, add recommendations, manage profiles.
- Admin Module: Manage user accounts, system monitoring, access control.
- Prediction System: Classify images using a trained CNN model.

❖ Non-Functional Requirements:

- Performance: Fast processing with real-time feedback.
- Security: Encrypted storage and SSL-based communication.
- Usability: Clean UI, responsive design, minimal training required.
- Portability: Compatible with Windows/Linux, major browsers.
- Maintainability: Modular codebase using open-source tools (Flask, TensorFlow).
- Scalability: Designed to support future integrations and larger datasets.

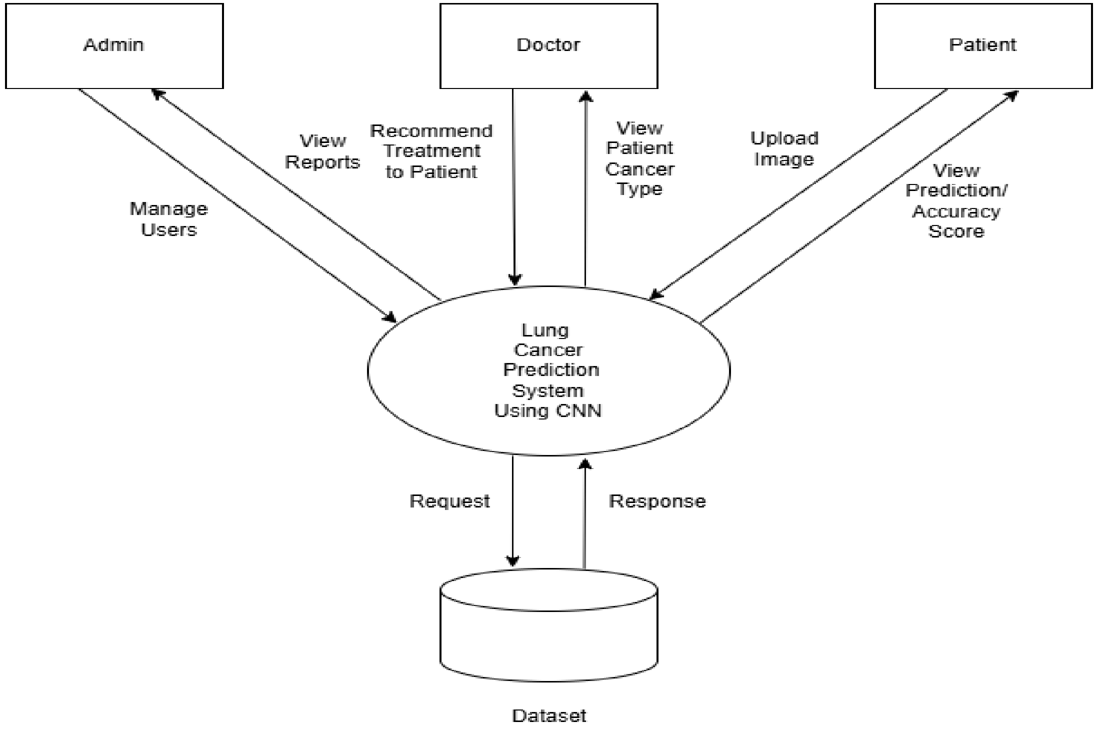
❖ Tech Stack:

Python, Flask, MySQL, TensorFlow, OpenCV, Bootstrap, FPDF

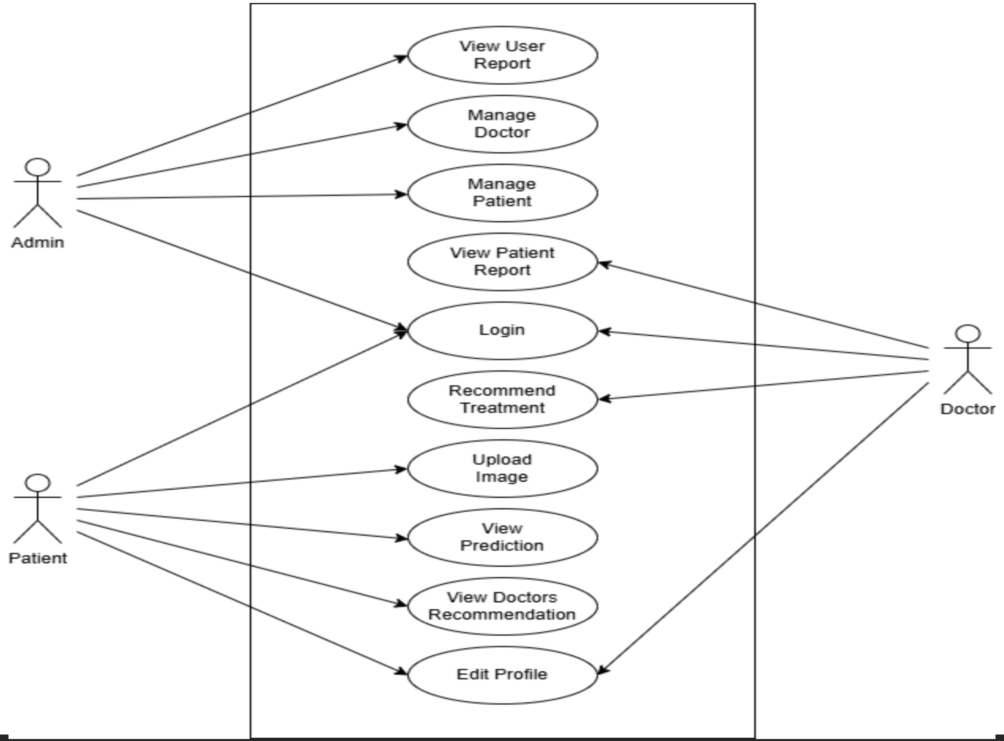
Design Documents

System design is a crucial stage in software development. It focuses on identifying the necessary modules, defining their functions, and outlining how they interact to deliver the intended outcome. This phase transforms user requirements into specific system features. It covers both the logical framework and the physical structure of the system. The logical design outlines elements like input files, output files, and databases, ensuring they align with project goals. Overall, this process helps clarify the structure and workflows needed to successfully build and implement the system.

Context Flow Diagram

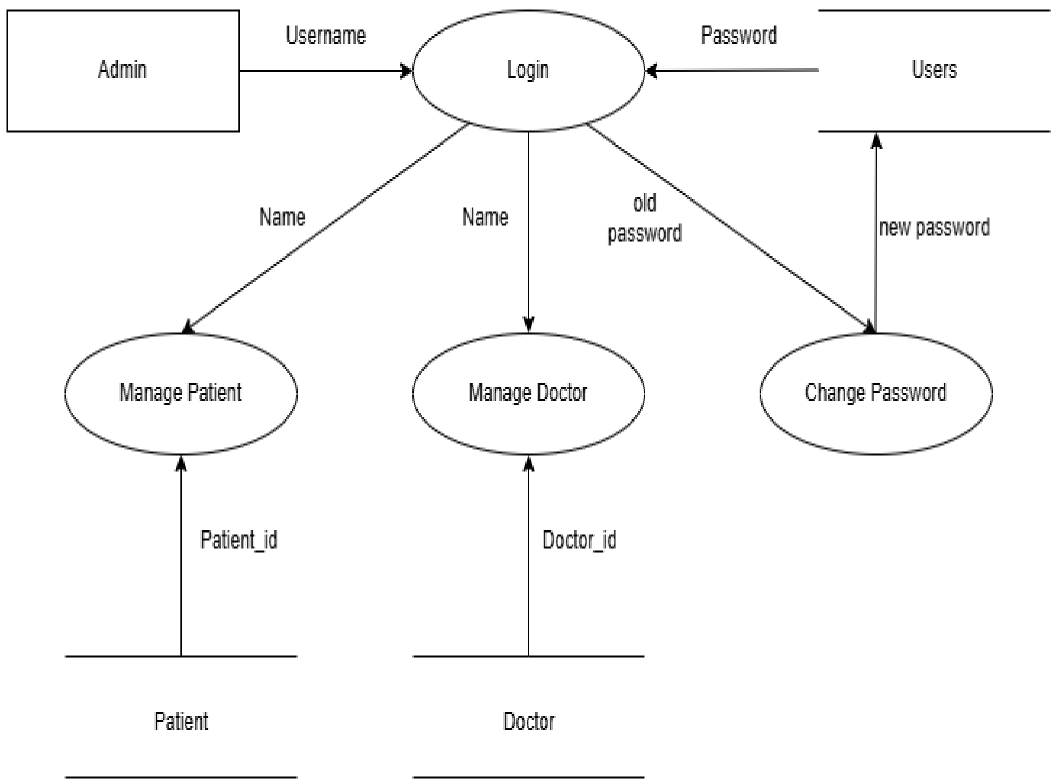


Use Case Diagram

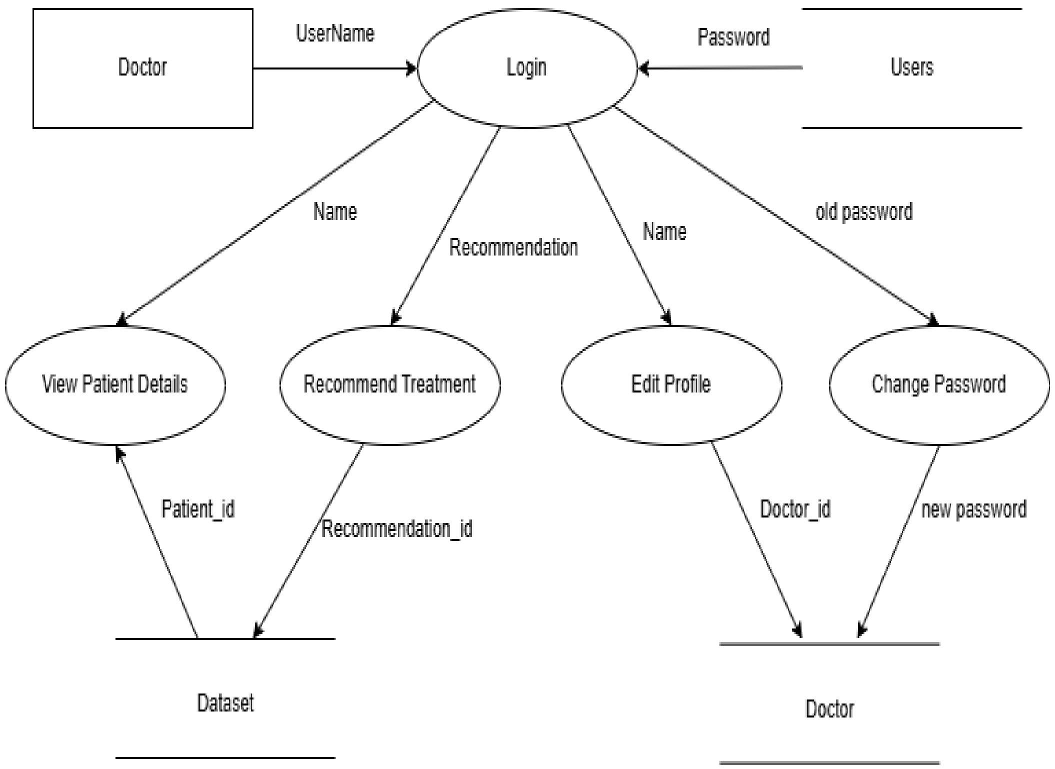


Design Documents

Admin DFD

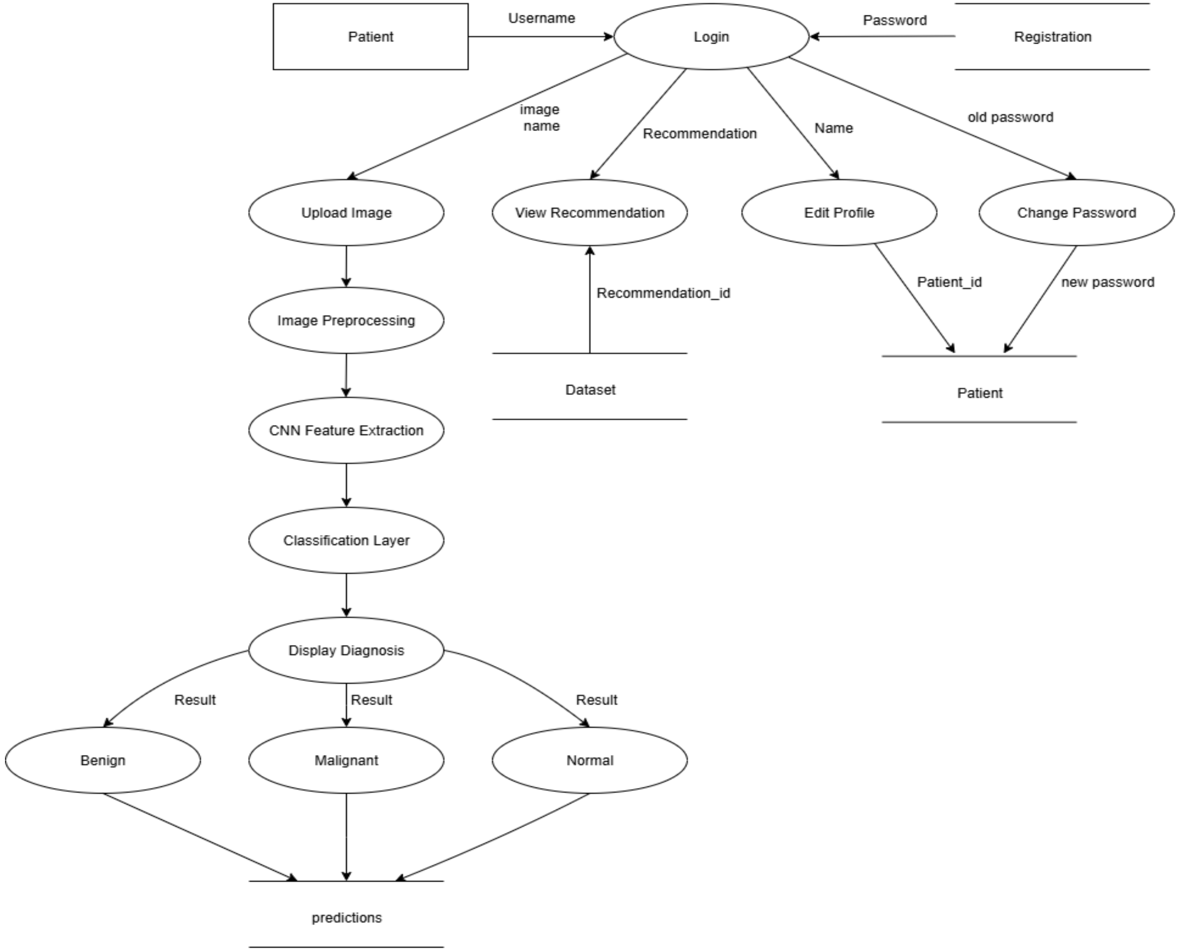


Doctor DFD

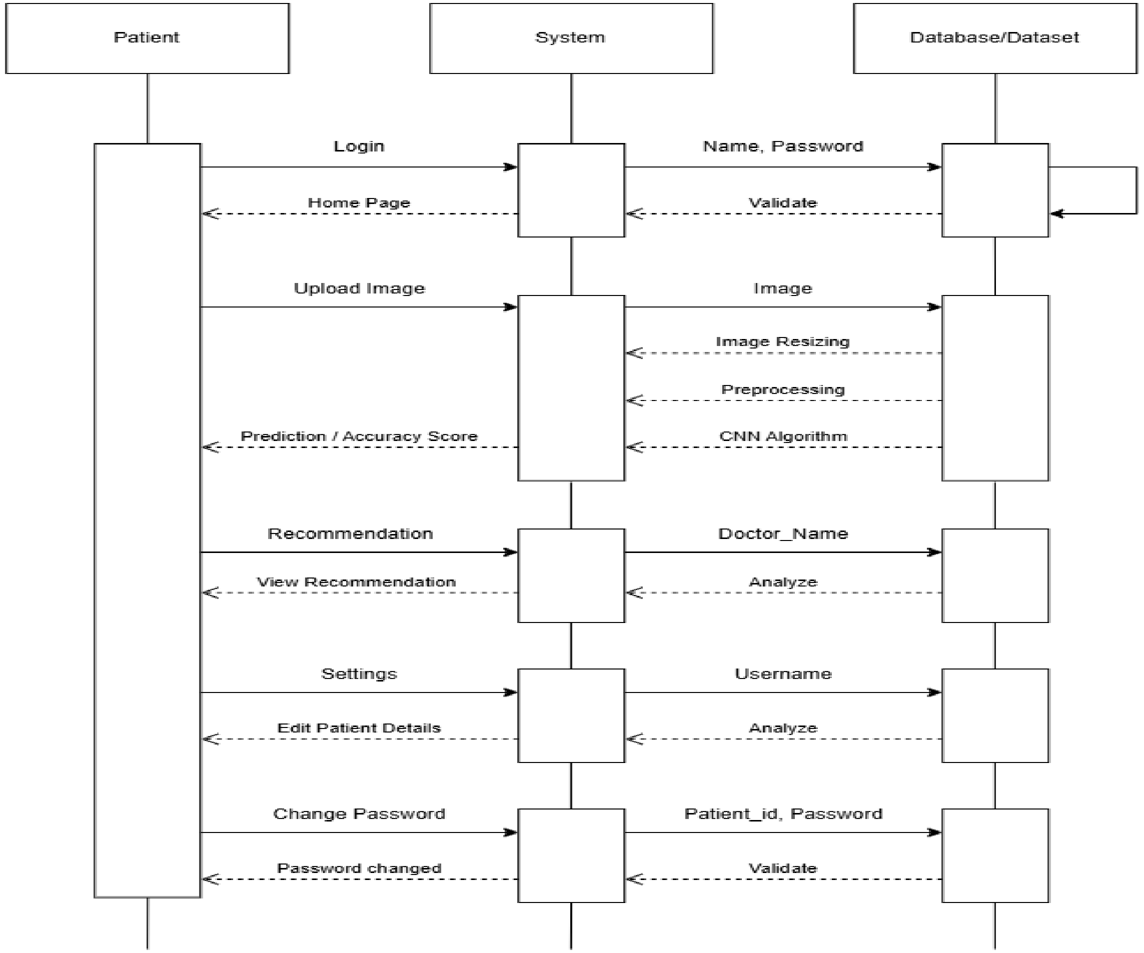


Design Documents

Patient DFD

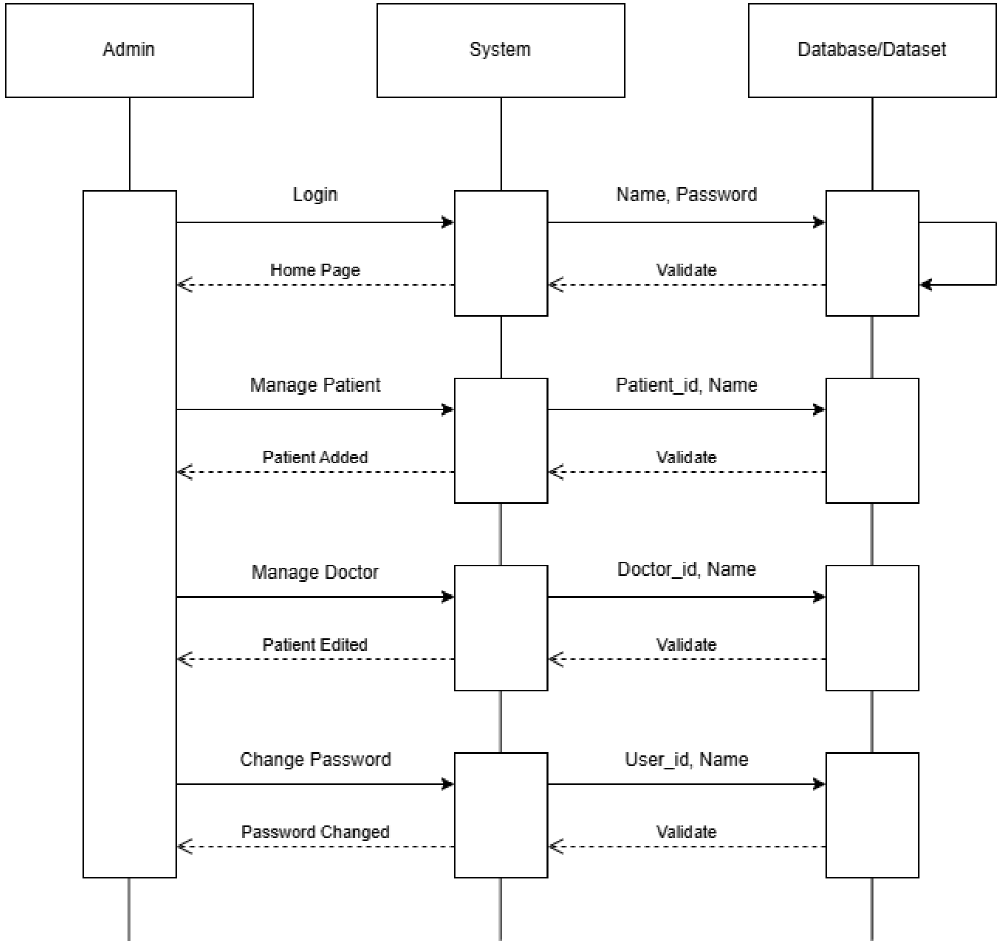


Patient Sequence Diagram

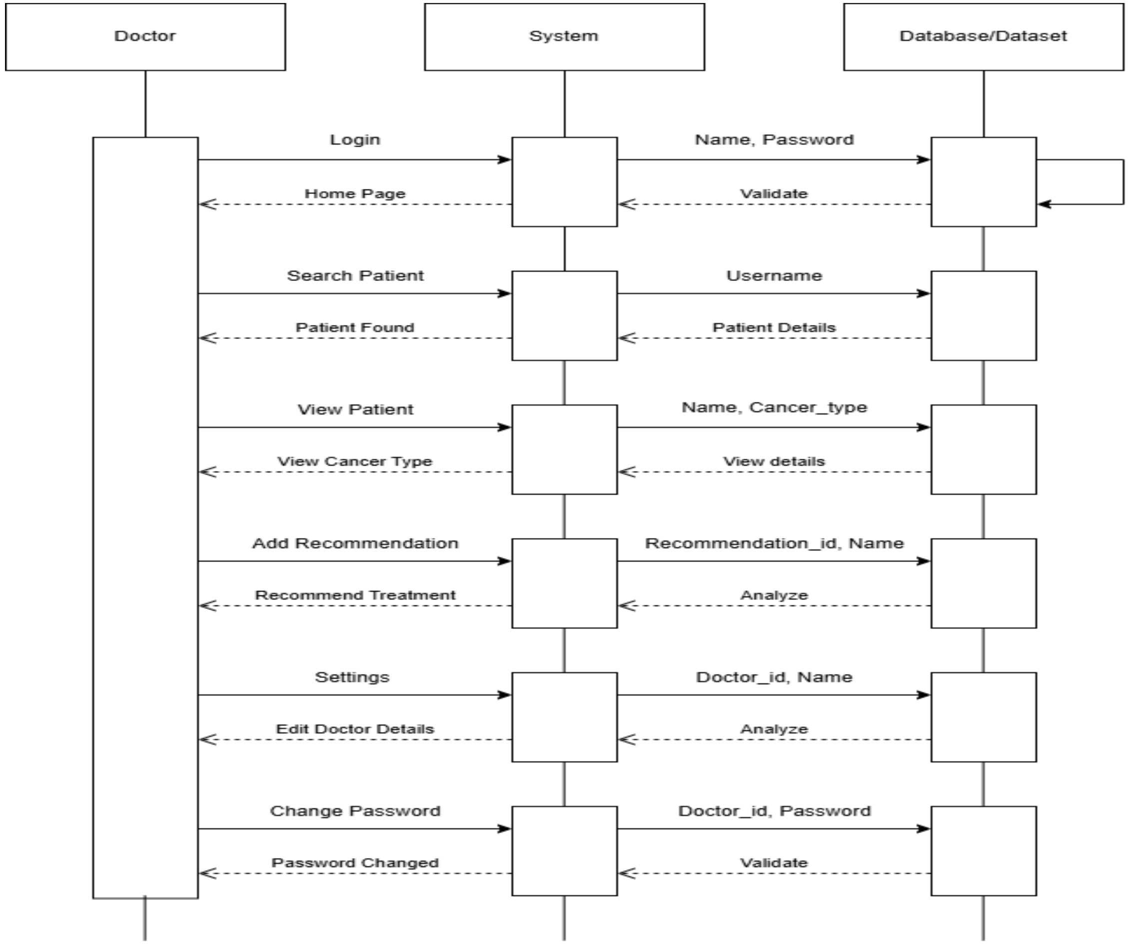


Design Documents

Admin Sequence diagram

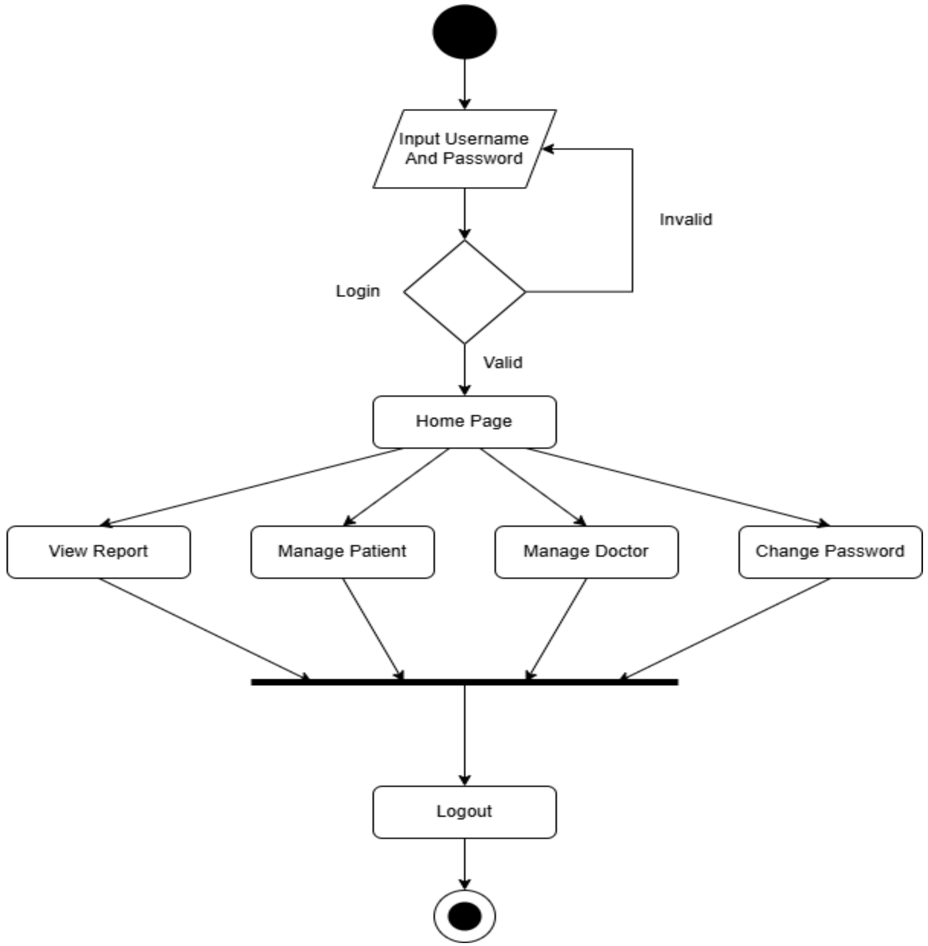


Doctor Sequence diagram

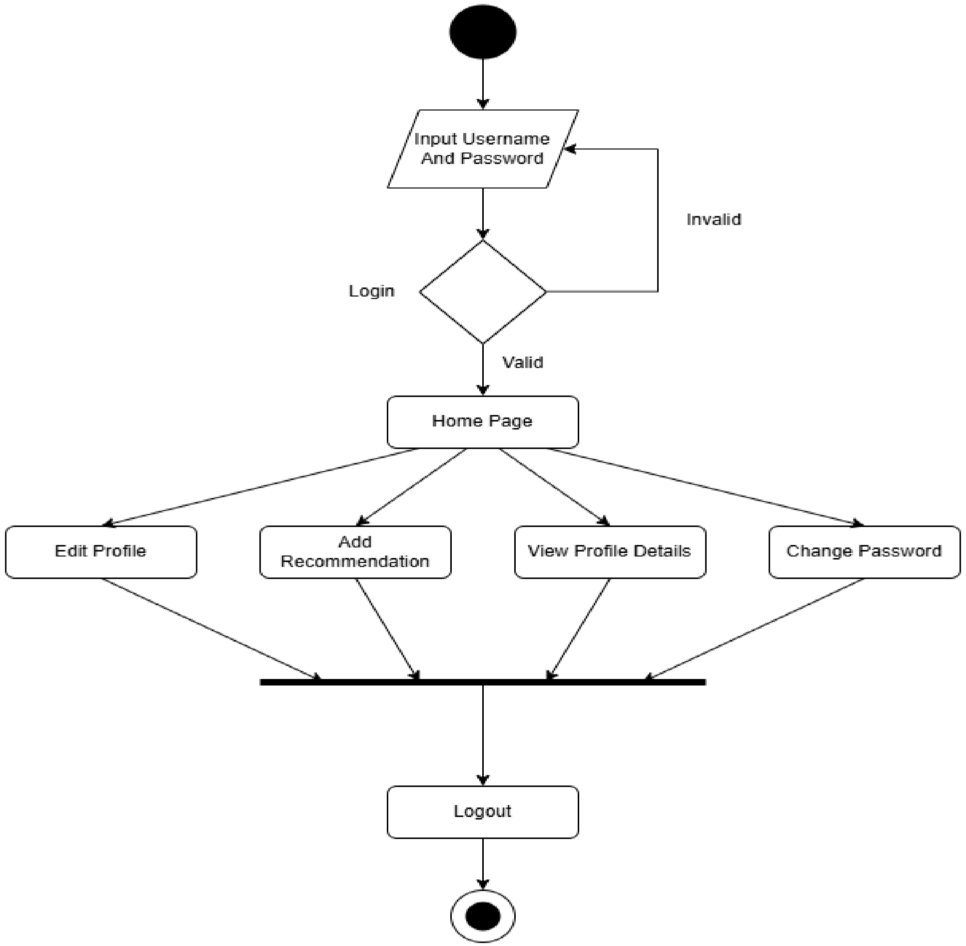


Design Documents

Admin activity diagram

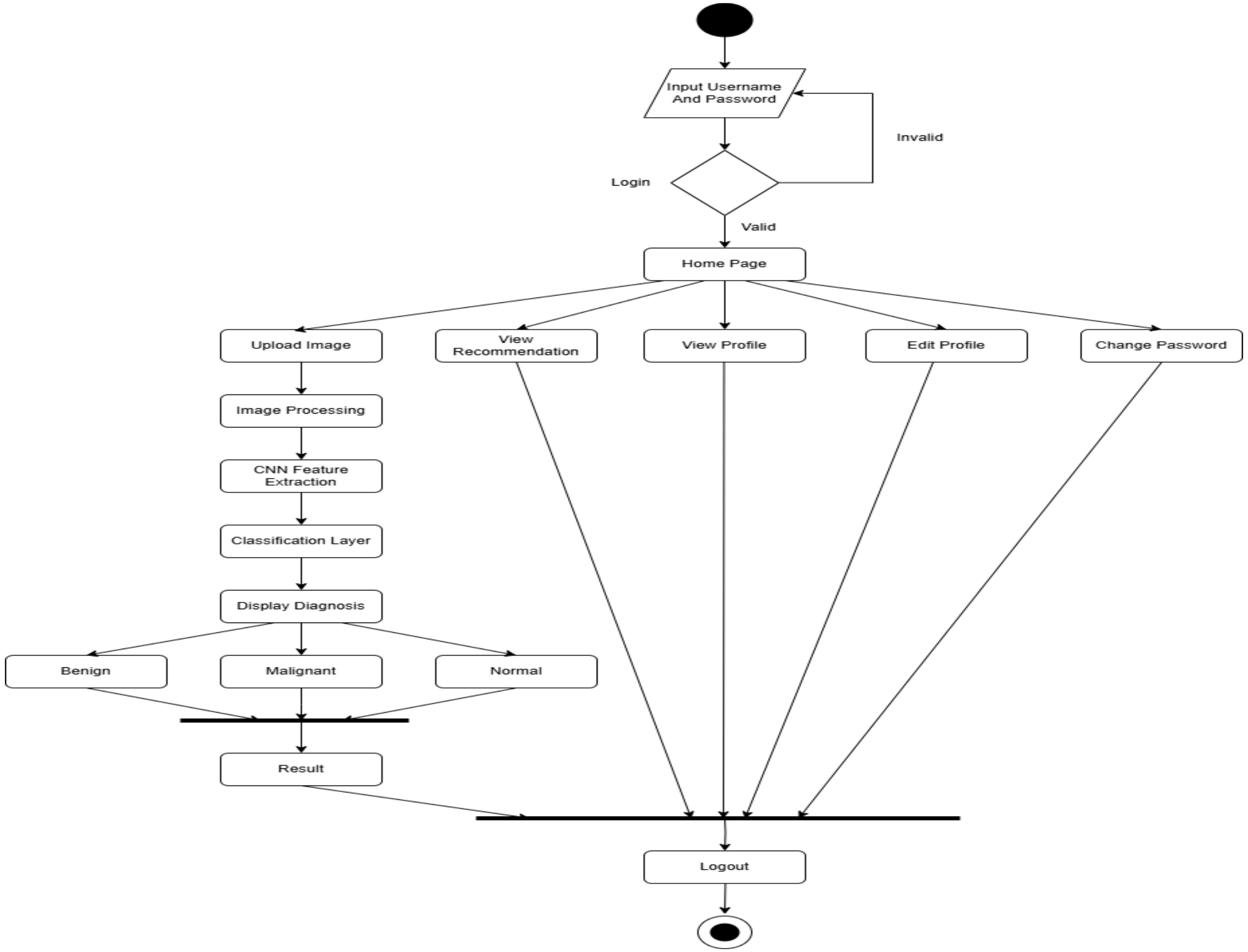


Doctor activity diagram



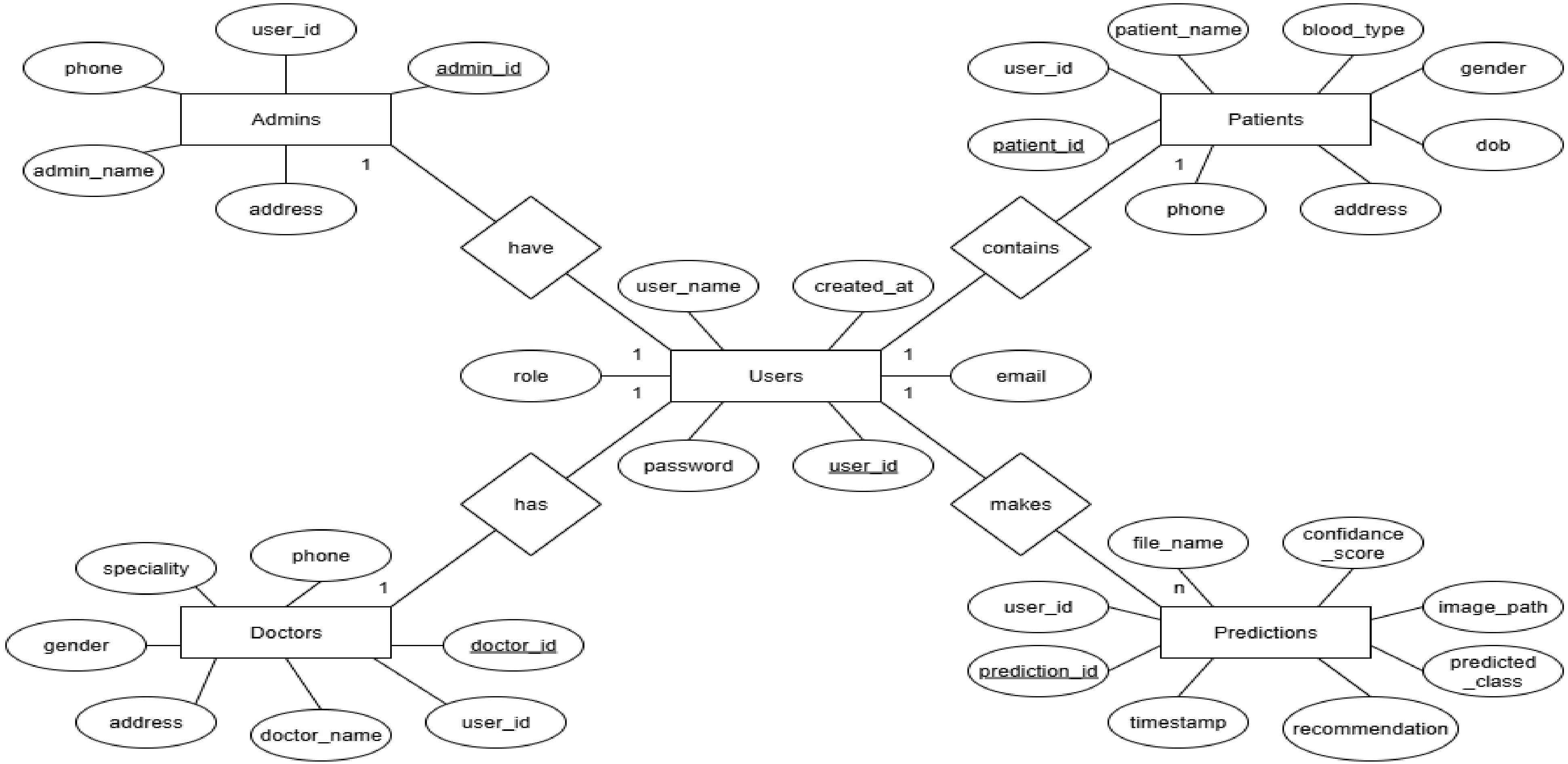
Design Documents

Patient activity diagram



Design Documents

ER - diagram



Testing and Implementation

Testing is a crucial procedure used to guarantee the accuracy, comprehensiveness, and caliber of software. By locating and fixing software bugs, it contributes significantly to a system's success.

TESTING FOR REGISTRATION

Tests case	Inputs	Tests descriptions	Output
1)	If any field left blank	All values must be entered.	All values must be entered.
2)	Username is Text	Valid username needs to be entered	Enter correct username
3)	If extension is .com For email id	Valid extension must be given	Only .com extension is allowed
4)	If Password is blank	Password is empty	Enter password
5)	Confirm password does not match	Password does not match	Password mismatch
6)	Valid input	Valid input	Registrations successful

TESTING IF INSERTION OF DATA IS VALID

Test case	Input	Test description	Output
1	Mandatory fields left empty	Mandatory fields cannot be left empty	Must enter data
2	Duplicate entry	Duplicate entry not allowed	Appropriate error message
3	Input without above faults	Valid input	Record inserted Successfully

Testing and Implementation

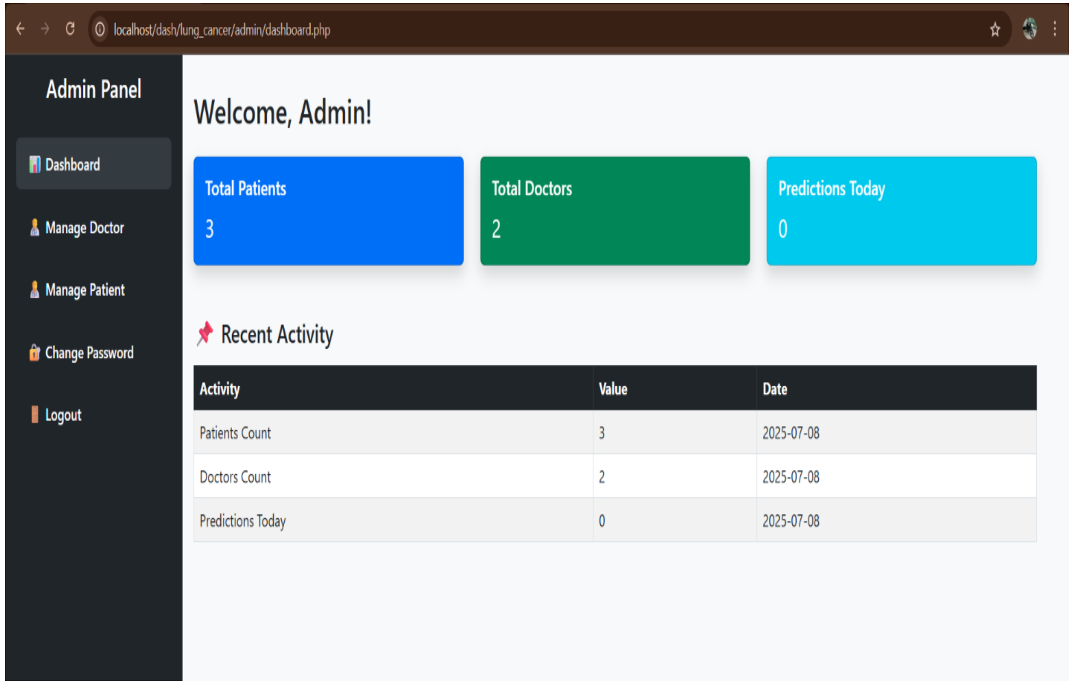
Executing a plan is the process of implementation. It involves an overview, a brief description, and the tasks required to carry out the implementation. During this phase, various test cases are evaluated and integrated into the system.

PSEUDO CODE FOR LUNG CANCER DETECTION

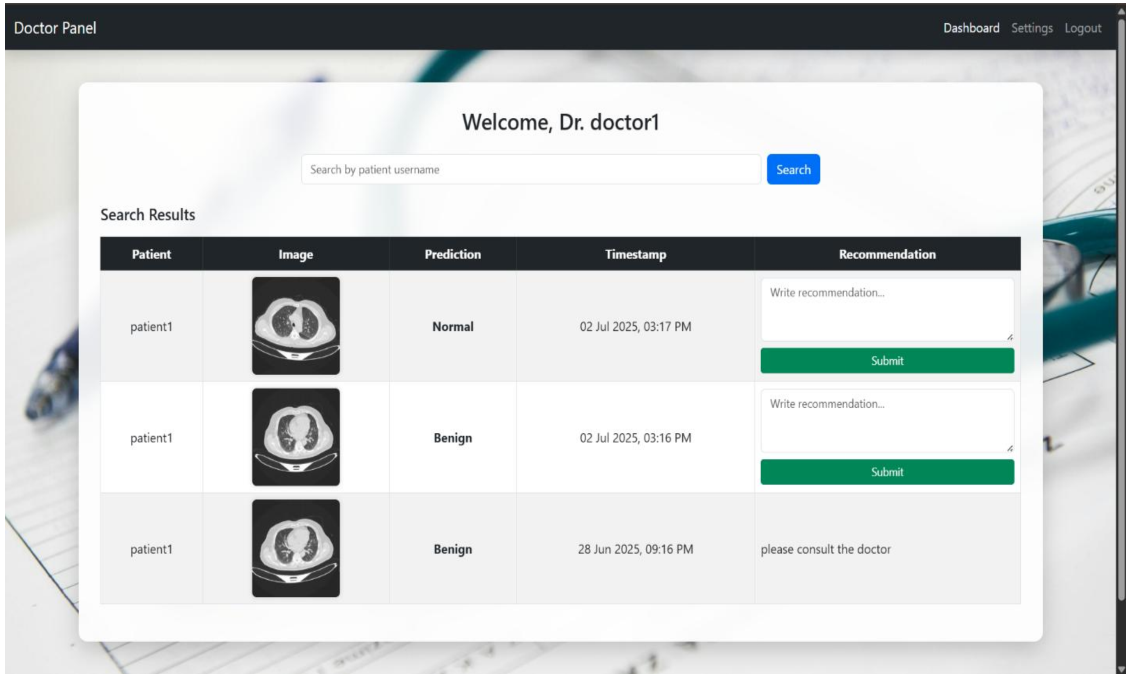
```
Begin
Input: High-resolution CT-Scan images
Preprocess the images:
    Apply normalization
    Perform noise reduction
    Apply contrast enhancement
    Standardize image size and resolution
Extract features:
    Identify texture, shape, density, and cancer size
Analyze using Convolutional Neural Network (CNN) features:
    Classify cancer as benign or malignant based on trained dataset
If cancer is malignant:
    Recommend further medical evaluations and treatment
Else if cancer is benign:
    Suggest routine monitoring and preventive measures
End
```

Screenshots

Admin Dashboard

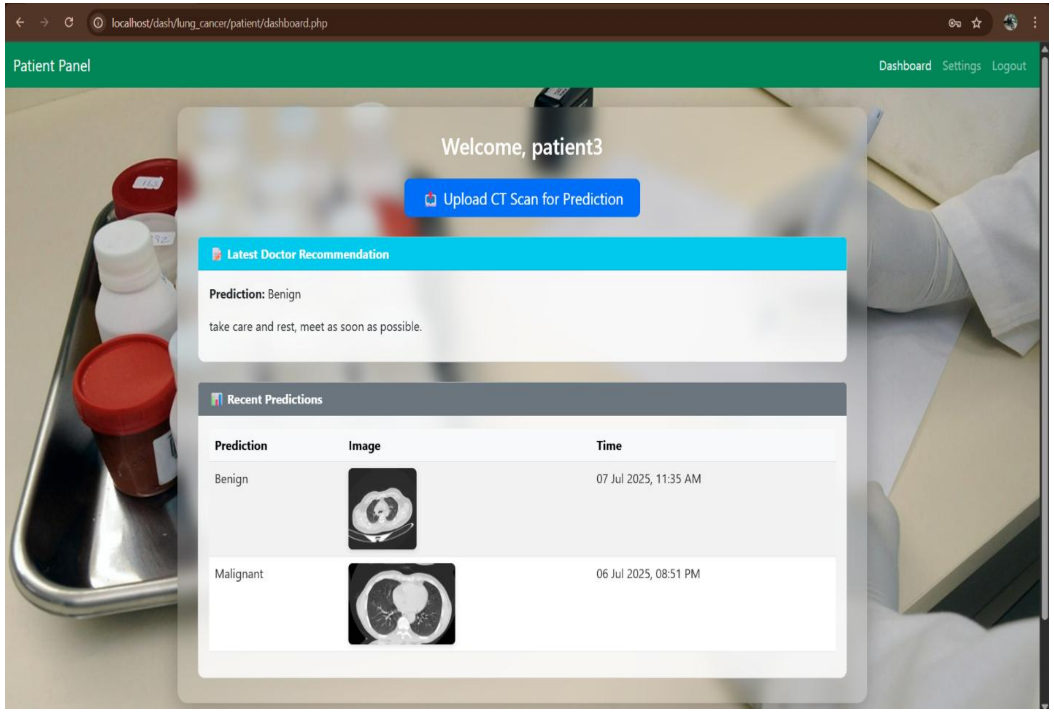


Doctor Home Page

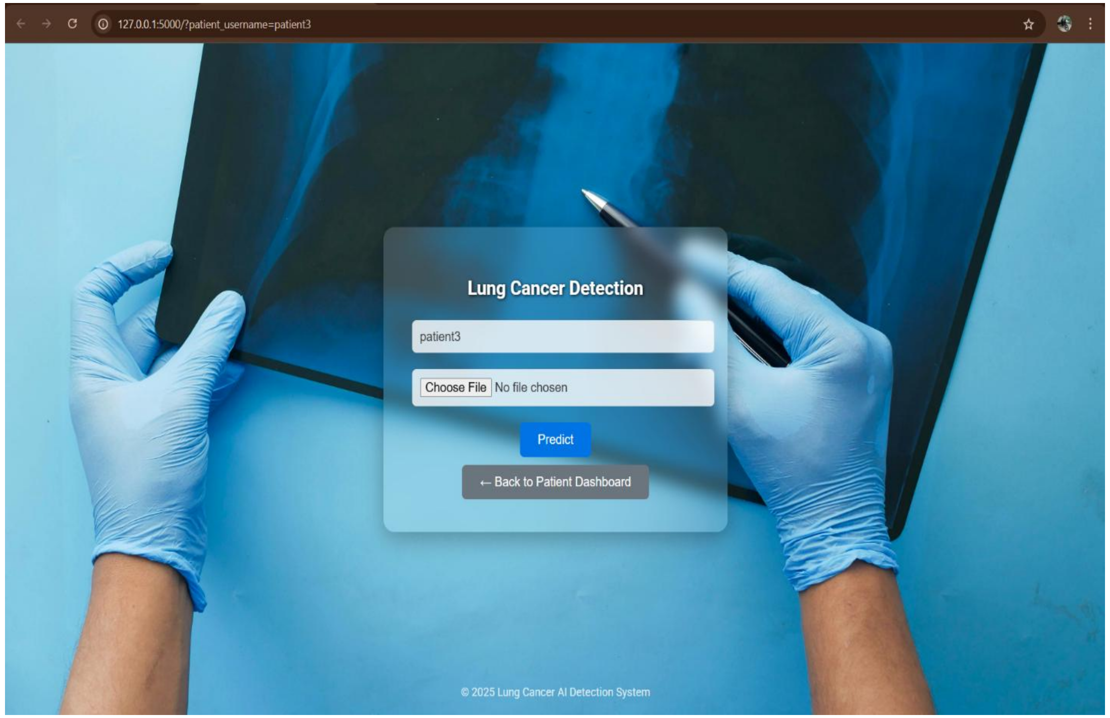


Screenshots

Patient Home Page

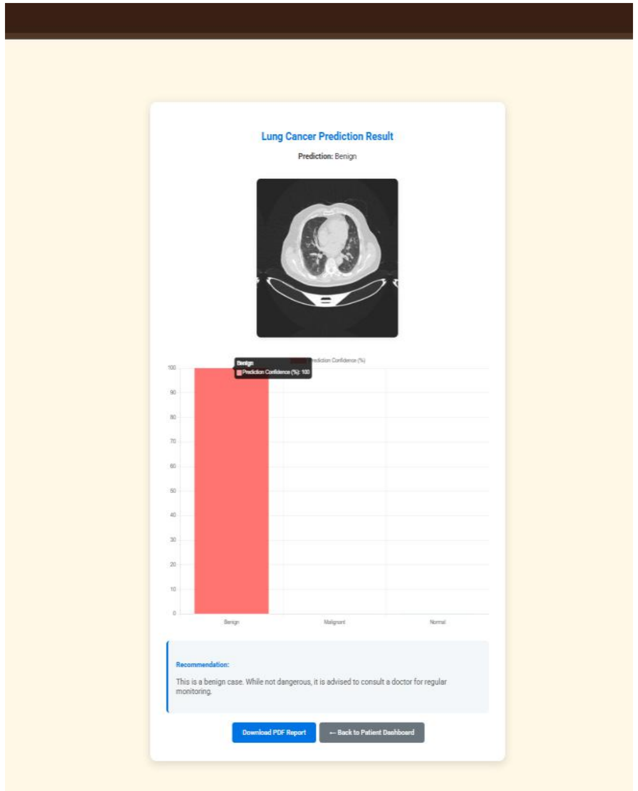


ML Prediction Page

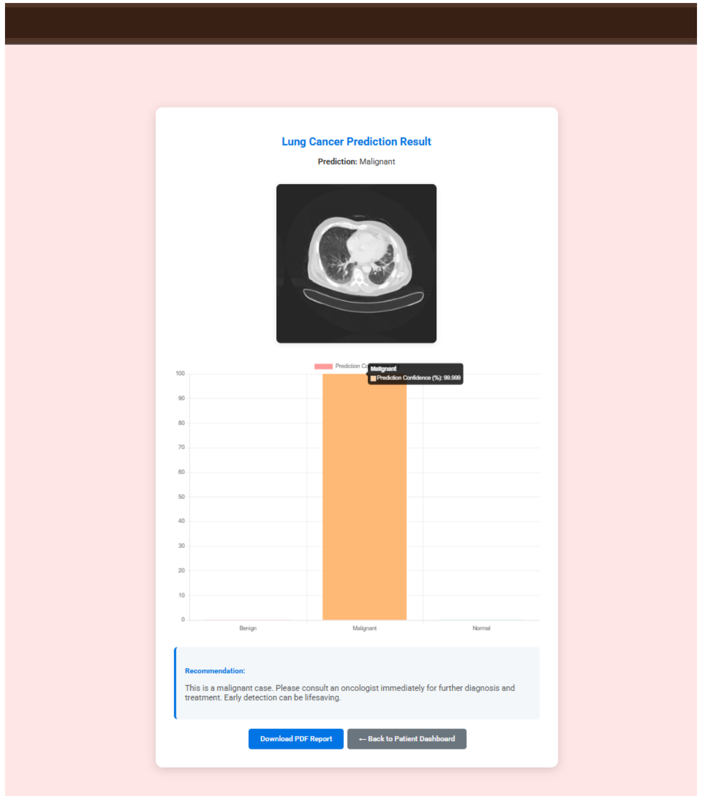


Screenshots

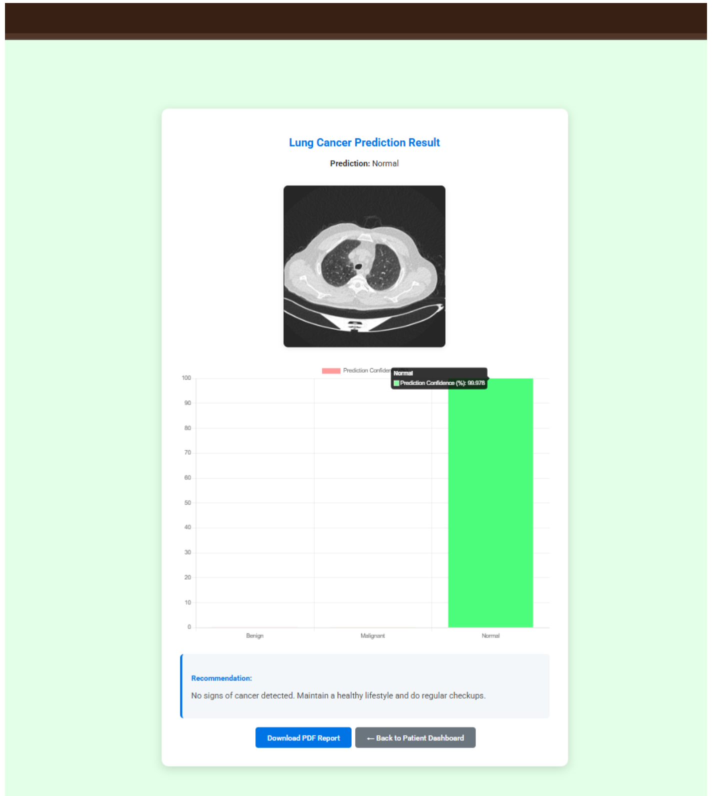
Benign Case Prediction



Malignant Case Prediction



Normal Case Prediction



Conclusion

- The Lung Cancer Detection project utilizing Convolutional Neural Networks (CNNs) marks significant progress in medical diagnostics by applying advanced deep learning techniques to classify CT scan images into benign, malignant, or normal categories. The CNN-based model addresses the shortcomings of traditional manual analysis and rule-based systems, offering a scalable and highly accurate approach for early detection of lung cancer.
- With a user-friendly interface and consistent performance across various datasets, the system improves the overall diagnostic process for doctors, patients, and healthcare staff. Its design integrates automated preprocessing, real-time predictions, and structured result generation, making it practical and dependable in clinical use.
- Beyond aiding in early detection, this solution highlights the potential of AI-driven healthcare tools. It demonstrates how technology can play a vital role in enabling faster decisions and more efficient patient care, contributing meaningfully to the fight against lung cancer.

Thank You